

Lab 1 – SpoilerAlert Product Description

Parker Davis

Old Dominion University

CS411W

Dr. Sarah Hosni

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1 Introduction

Food recalls are constant occurrences across the United States, posing significant risks to public health. Food is a daily necessity for everybody, therefore the consequences of an uncaught disease or allergen contamination are great. Despite this, there is not an adequate reporting system to keep people informed of these recalls. Of particular risk are children, who may be fed recalled food with risks that their parents are unaware of. Often, there are unmarked allergens, allergen cross-contamination (University of Nebraska-Lincoln), or even harmful diseases such as *Listeria* (Food Safety and Inspection Service, 2023). These diseases are especially serious for those with immunodeficiencies. Food recalls also come with economic consequences. Companies may take significant losses both economically and to their reputation, as seen in the case of ConAgra's peanut butter recall (US Justice Department, 2016). For these reasons, information reaching consumers fast is vital.

Before anybody can become better at paying attention to recalls, they must first be aware that recalls are happening in the first place. Most people do not hear about the vast majority of recalls. Only 13% of Americans check government websites for recalls, and only 3% are subscribed for automatic reports (Hallman, 2025). Then there is the worry about how important any one recall is to a person. Notification fatigue may arise when a user gets constant notifications about recalls that would never affect them. This could cause users to stop paying attention to notifications, or turn them off completely. What is needed is a service that can provide users with relevant information promptly.

Such a service would need to supply official government data from the FDA to keep users well informed. It would need to allow users to customize their experience as best fits them,

so that they do not shy away from using the service. An ideal service for this issue would also allow users to only receive recall alerts that relate to them and their personal purchases.

SpoilerAlert will be an application that fulfills all these requirements. It will deliver fast, reliable, and customizable recall notifications to users. This is so that consumers can avoid the health risks that come with recalled food. Through the use of the FDA's recall API, SpoilerAlert will alert users of any and all recalls as soon as possible. With barcode and receipt scanning, users can both check that what they have bought is safe and can be alerted about any future recalls that may affect their food. Previous purchases will be securely saved so that if a recall has the potential to affect a user, they will be notified before consumption.

2 Product Description

SpoilerAlert will be a phone application that will monitor official FDA recall sources, match these recalls with the user's purchases, and alert users of relevant recalls through their phone's built in notification system. This system will be built upon three application layers: presentation, application, and data.

2.1 Key Product Features and Capabilities

SpoilerAlert's primary purpose is to gather recall data and notify the application's users about said data. The customization of this data is another focus. All recall and product information is able to be filtered and searched from the app. All purchase information is also saved inside the inventory, so that the user may be notified only of recalls that relate to them.

Recall data will be sourced from the government's channels that announce them in the first place. Both of the agencies that are concerned with food recalls, the FDA and CDC, have APIs that allow for sourcing of recall information. This data will be used to link existing products in SpoilerAlert's database. So that users will be alerted when a recall may affect them.

SpoilerAlert's database will be filled with product information. If a certain product is not stored within the database, then the application will check Barcode Lookup so that it can keep itself updated with all needed information. Barcode Lookup has an API that allows SpoilerAlert to acquire product information with only a UPC. The intent here is to require as little human input as possible so that information can be updated before the user notices any slow down. The UPCs that SpoilerAlert uses to find and product data are sourced in two ways. First, SpoilerAlert will always try to extract a UPC, or enough product information to find one, from all recalls acquired through the FDA recall API.

The second method for SpoilerAlert to obtain a UPC is through any user scanning a product's barcode. SpoilerAlert will access a user's camera on their phone to scan a barcode placed in its view. SpoilerAlert will then look that UPC up in the database of products. If the product is not found there, then the application will attempt to use the Barcode Lookup API to find and register the product. Once registered, the product may be saved in the user's inventory.

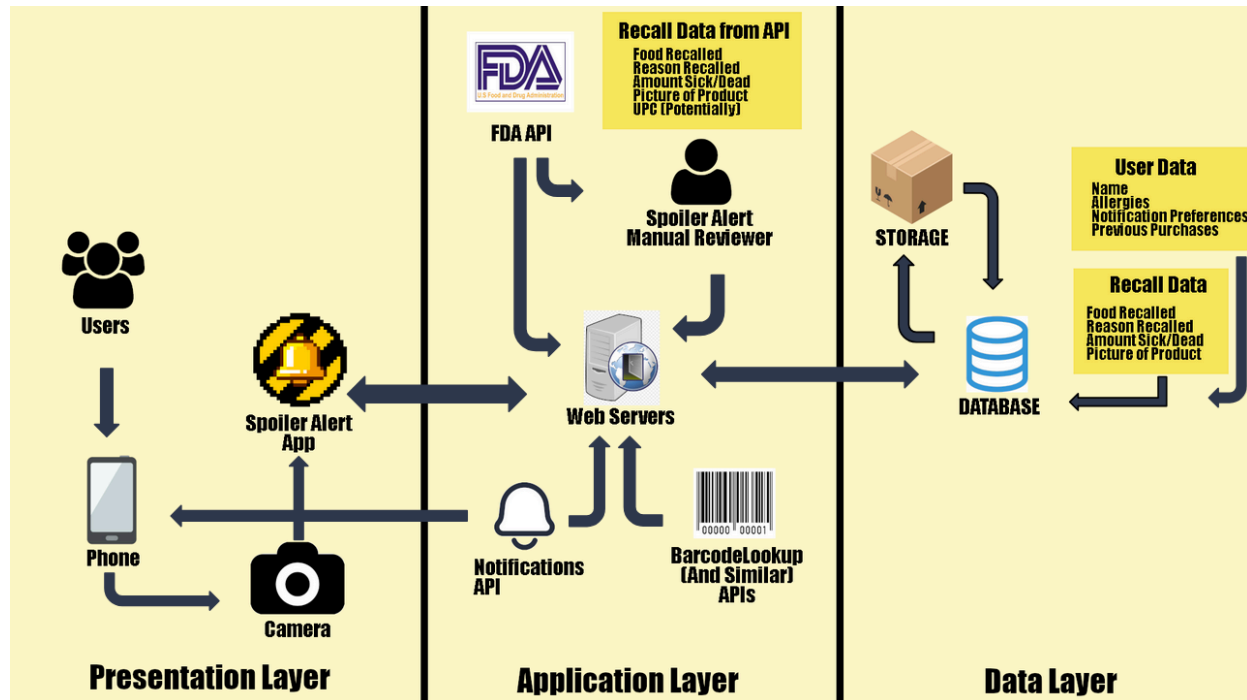
A user's inventory acts as a purchase history for the user, allowing them to see what was bought in the past as well as present information about the product to them. Any recalls the product was involved in will also be displayed here. The inventory will contain both a search function to find products easily as well as filtering so that a user may find all products fitting a given criteria. The inventory will also display scanned receipts. By scanning a receipt, the user can save and track an entire purchase with one scan.

Both the search and filtering functions are shared by the home page of SpoilerAlert. This is where recent recalls and news reports on recalls and food safety are displayed. The search function here will allow users to find information on any product they may be interested in. The

filtering function will allow users to find information on a category of products, filter out a category, or filter out news reports entirely.

2.2 Major Components (Hardware/Software)

SpoilerAlert will have three layers that data moves between. The presentation layer is found on any user's device. It contains the phone application, phone peripherals, and the user themselves. The phone application communicates with the application layer. The application layer contains the servers that SpoilerAlert runs off, it is the servers that communicate with the various APIs that are used, as well as moves data to and from the database. The database sits on the data layer, which holds all the product and recall data that SpoilerAlert collects and uses. This relationship is shown below in Figure 1.

Figure 1*Major Functional Components Diagram*

The SpoilerAlert app will be the point where the user interacts with the entire SpoilerAlert system. It will display the user's relevant recalls, purchase history, and news feed. Data received from the database through the servers. The app is responsible for utilizing the camera of whatever system it is on to scan barcodes and receipts. It will also be giving the user any mobile notifications that they need to receive.

The servers inside the application layer are what gather, parse, and manage all the data that SpoilerAlert requires to function. The primary source of recall data will be the FDA's own openFDA API. This API provides data about current and past recalls in a JSON format. This allows the SpoilerAlert servers to easily parse recall information (openFDA). Product information will be sourced through Barcode Lookup's API, providing all needed product

information in the same JSON format (Barcode Lookup). Once a product's information is found and matched, it will be added to the database on the data layer.

The data layer contains all the data of SpoilerAlert. User, product, and recall information is all stored there. The database will be run on MySQL, and will be queried from the SpoilerAlert servers on the application layer.

3 Identification of Case Study

4 Glossary

Recall: When a manufacturer notifies the market of a product with a defect or potential danger.

UPC: Universal Product Code. It is a number that uniquely identifies anything you can buy from a store.

Barcode: A visual representation of a UPC that is used at a store to scan a product so you may purchase it.

FDA: The Food and Drug Administration. A federal agency in the USA that works to make food products safer.

CDC: Centers for Disease Control and Prevention. National public health agency of the USA.

USDA: United States Department of Agriculture. Agricultural department of the United States government.

JSON: A file format that allows for data to be stored in a human readable format that is consistent and easy to work with in computer code.

API: Application Programming Interface. It allows programs to communicate with each other. In this case, it allows SpoilerAlert to quickly acquire data from official sources.

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